Name: ______________________

Worksheet

Calculating theoretical probabilities from a table (one event)

Task A: Finding probabilities from a frequency table

1) The table shows the number of sweets of each flavour in a novelty box of sweets.

flavour	frequency
strawberry	11
caramel	10
toffee	8
fudge	12
orange	8
chocolate	9
grass	8
earwax	3
fish	5
garlic	6
total	80

a) Find the probabilities for picking each of the following flavours when a sweet is chosen at random.

P(strawberry) =　　　　P(caramel) =

P(toffee) =　　　　P(chocolate) =

P(grass) =　　　　P(earwax) =

P(fish) =　　　　P(garlic) =

b) The nasty flavours are grass, earwax, fish and garlic. What is the probability of choosing a nasty flavoured sweet?

2) Sam is carrying out a data investigation about how pupils in a nearby village travel to school. She conducts a survey and records her results in a tally chart.

a) Complete the frequency column of the table.

mode of transport	tally	frequency
car	𝍸 𝍸 𝍸 𝍸	
bus	𝍸 𝍸 𝍷𝍷	
bicycle	𝍸 𝍷𝍷𝍷𝍷	
walk	𝍸 𝍸 𝍸 𝍷	

b) To further her investigation, Sam chooses a pupil at random to interview.

i) What is the probability that the person chosen travels by car?

ii) What is the probability that the person chosen travels by bus?

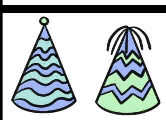
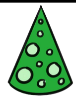
iii) What is the probability that the person chosen travels by bicycle?

iv) What is the probability that the person chosen walks to school?

Task B: Finding probabilities from an outcome table

1) Party hats from a pack are sorted according to how they are decorated.
A hat is chosen at random from the pack.

Find each of the following probabilities.

	stripes	no stripes
topper		
no topper		

a) P(stripes and topper) =

b) P(stripes and no topper) =

c) P(no stripes and topper) =

d) P(no stripes and no topper) =

e) P(stripes) =

f) P(topper) =

2) A deck of cards contains the numbers 1 to 20.

a) Sort the numbers 1 to 20 into the table.

b) A card is chosen at random from the deck.
Find the probabilities of the following outcomes.

i) P(multiple of 4 and factor of 20) =

ii) P(multiple of 4 and not factor of 20) =

iii) P(multiple of 4) =

iv) P(not multiple of 4) =

	multiple of 4	not multiple of 4
factor of 20		
not factor of 20		